

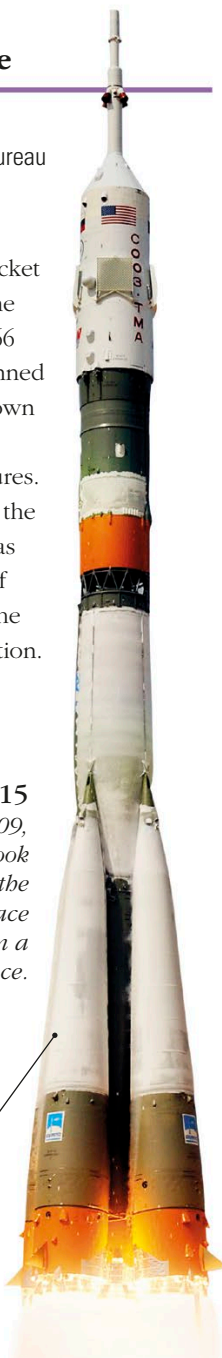
Space workhorse

- **What?** Soyuz rocket
- **Who?** OKB-1 design bureau
- **Where and when?** Russia, 1966

The most successful rocket in the world today is the Soyuz. In use since 1966 for manned and unmanned launches, Soyuz has flown more than 1,700 times, with relatively few failures. Since the retirement of the Space Shuttle, Soyuz has been the only means of sending astronauts to the International Space Station.

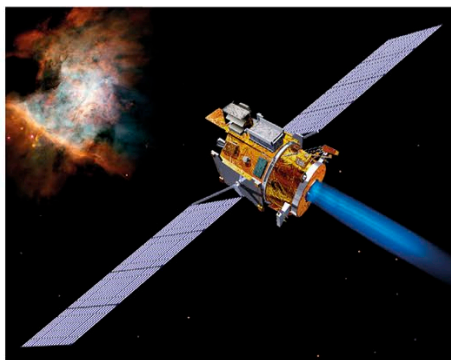
► **SOYUZ TMA-15**
Blasting off in 2009, this Soyuz rocket took three astronauts to the International Space Station to begin a six-month stay in space.

The lower stage consists of four booster rockets clustered around the core rocket.



FAST FACTS

- The first flight of Robert Goddard's liquid-fueled rocket in 1926 lasted about 20 seconds and reached a height of just 41 ft (12.5 m).
- The first stage of a Saturn V burns 4.7 million lb (2.1 million kg) of fuel in just 161 seconds, boosting the rocket to a speed of 6,164 mph (9,920 km/h).
- Ion engines use their fuel 10 times more efficiently than chemical rockets, which means they require far less fuel to lift a rocket off the ground and into orbit.



Ion engines

- **What?** Deep Space 1
- **Who?** NASA
- **Where and when?** US, 1998

Most rockets burn fuel explosively to generate thrust and overcome gravity. Ion engines use electricity from solar cells to eject a stream of charged particles (ions). The thrust from an ion engine is tiny, but operating over months rather than seconds they can achieve tremendous speeds. NASA first tested ion propulsion on its robotic space probe, Deep Space 1.

Rocket-powered aircraft

- **What?** SpaceShipOne
- **Who?** Burt Rutan, Scaled Composites LLC
- **Where and when?** US, 2004

Carrier aircraft have transported small rocket-powered vehicles for release at high altitude since the 1950s. American engineer Burt Rutan's SpaceShipOne was the world's first private, rocket-powered, manned spaceship. Adopted by Virgin Galactic, its wing design allows it to float slowly back to Earth after briefly reaching space without going into orbit around Earth.



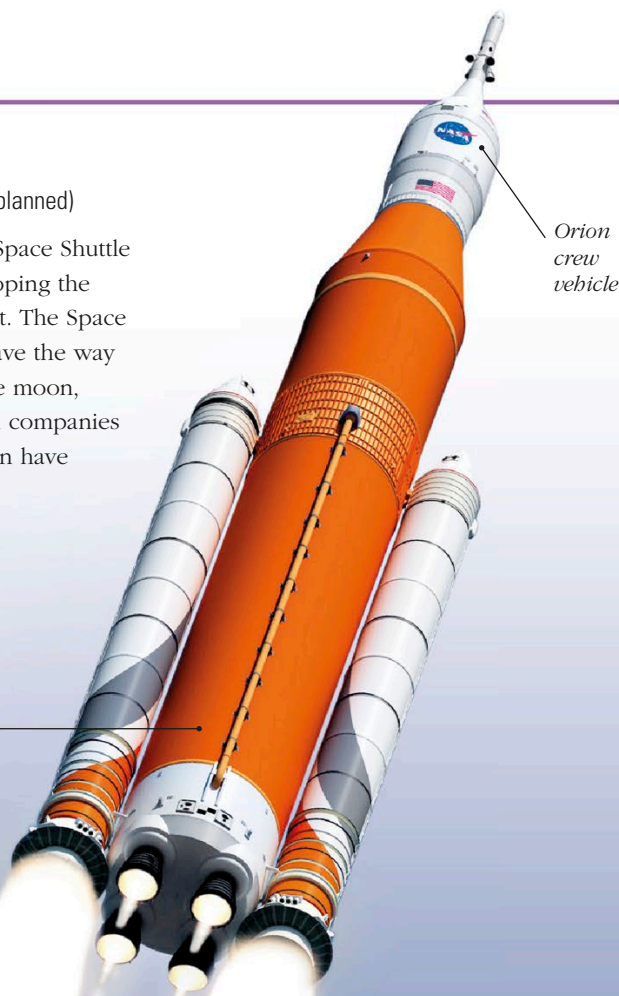
Future rockets

- **What?** SLS "Block 1"
- **Who?** NASA
- **Where and when?** US, 2019 (planned)

Since its decision to retire the Space Shuttle in 2011, NASA has been developing the most powerful rocket ever built. The Space Launch System (SLS) should pave the way for new crewed missions to the moon, Mars, and beyond. Commercial companies such as SpaceX and Blue Origin have equally ambitious plans.

SLS will be taller than the Statue of Liberty and produce 30 times more thrust than a 747 jet.

Illustration of Space Launch System in flight



Orion crew vehicle